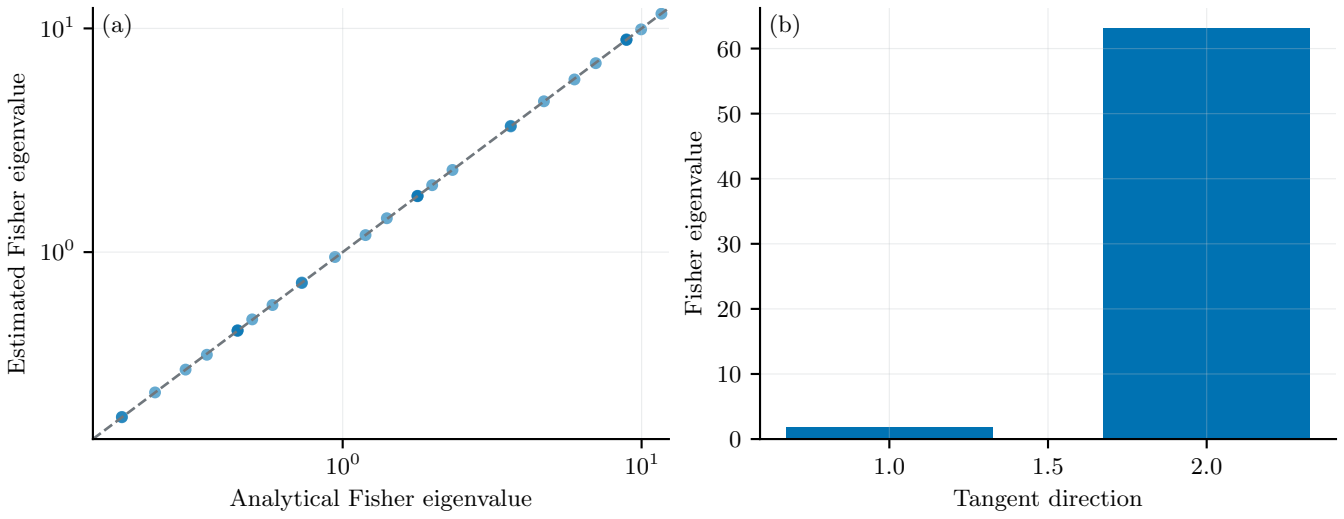
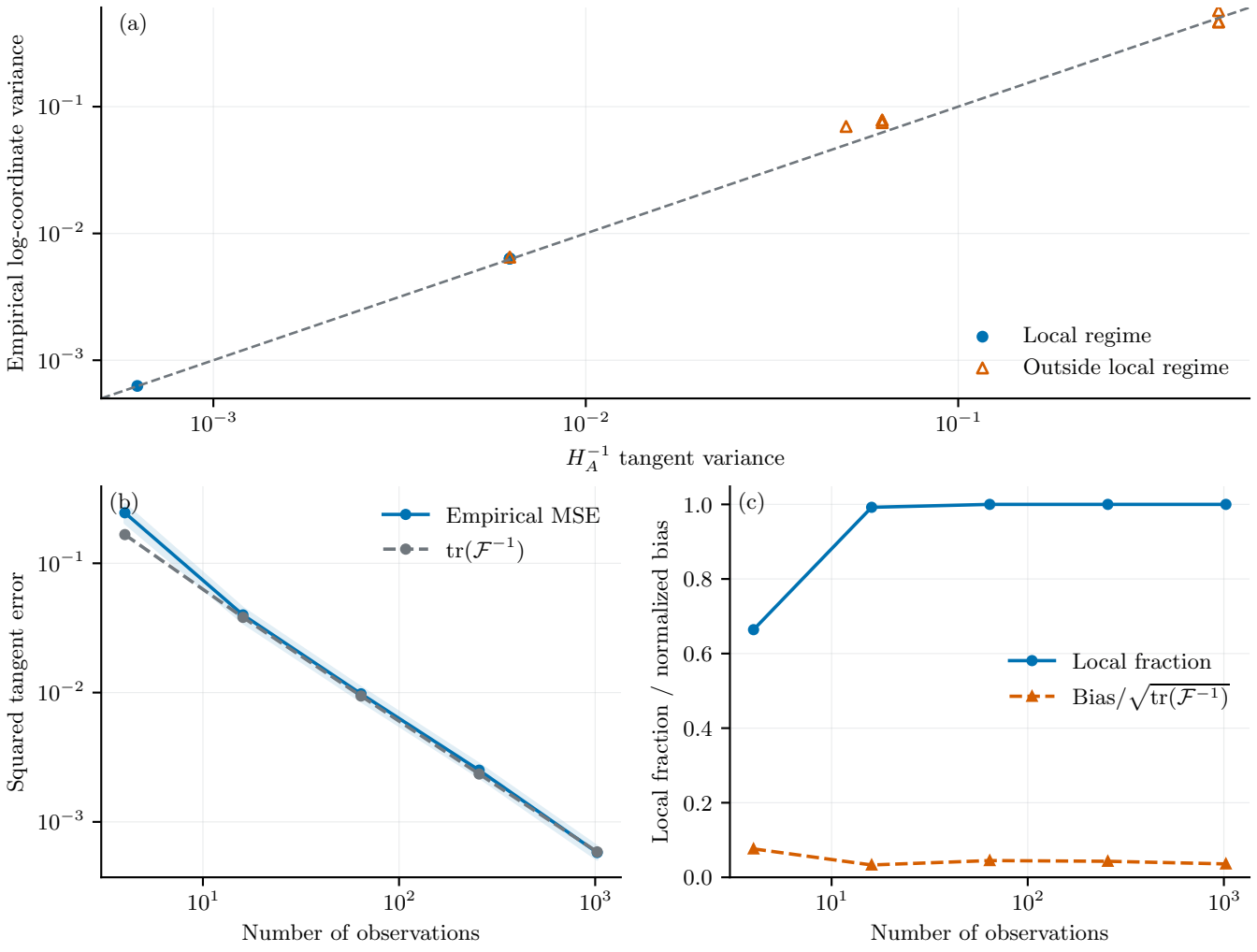
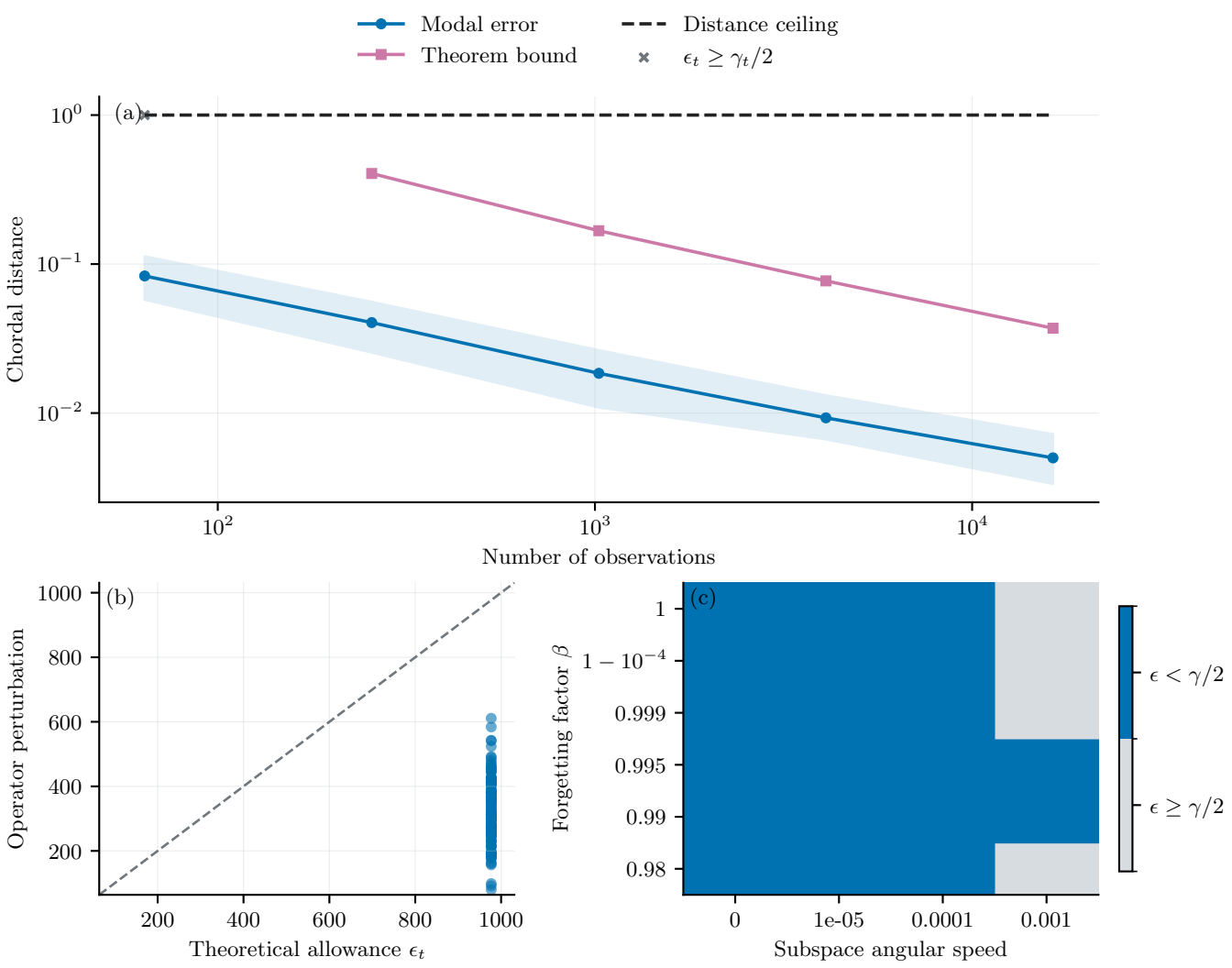
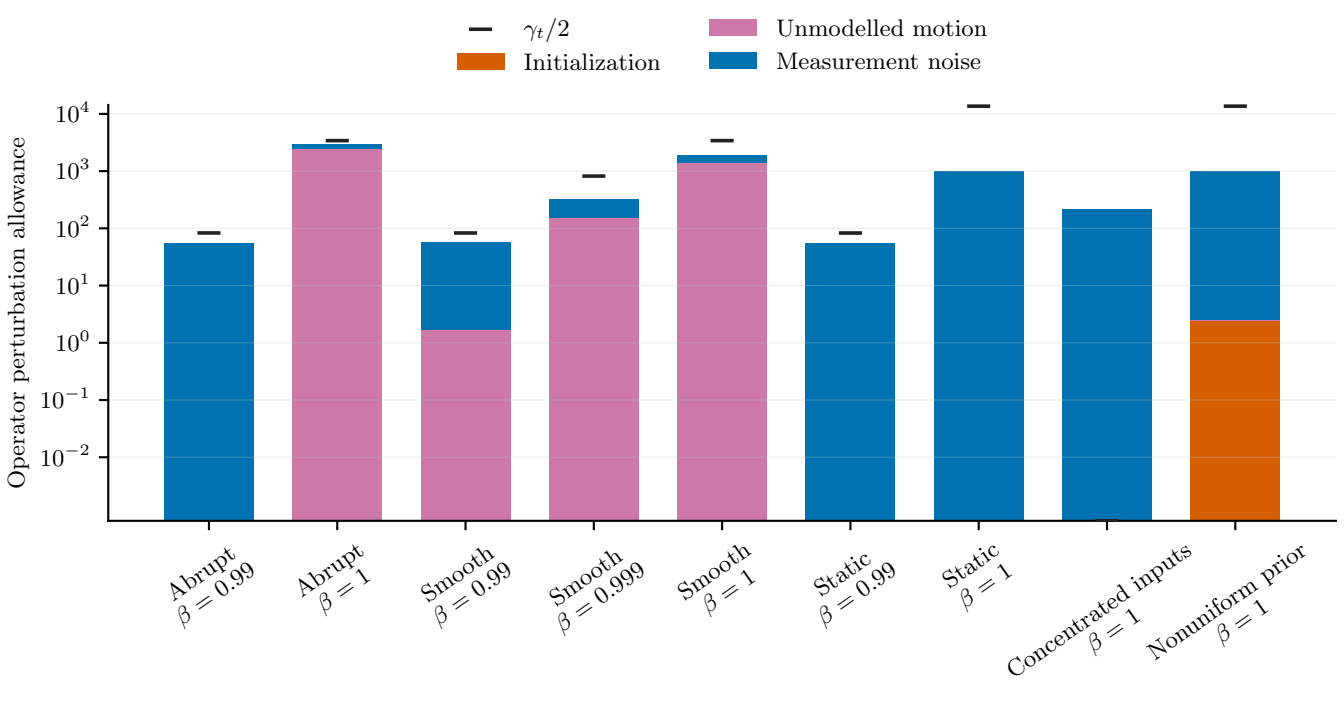
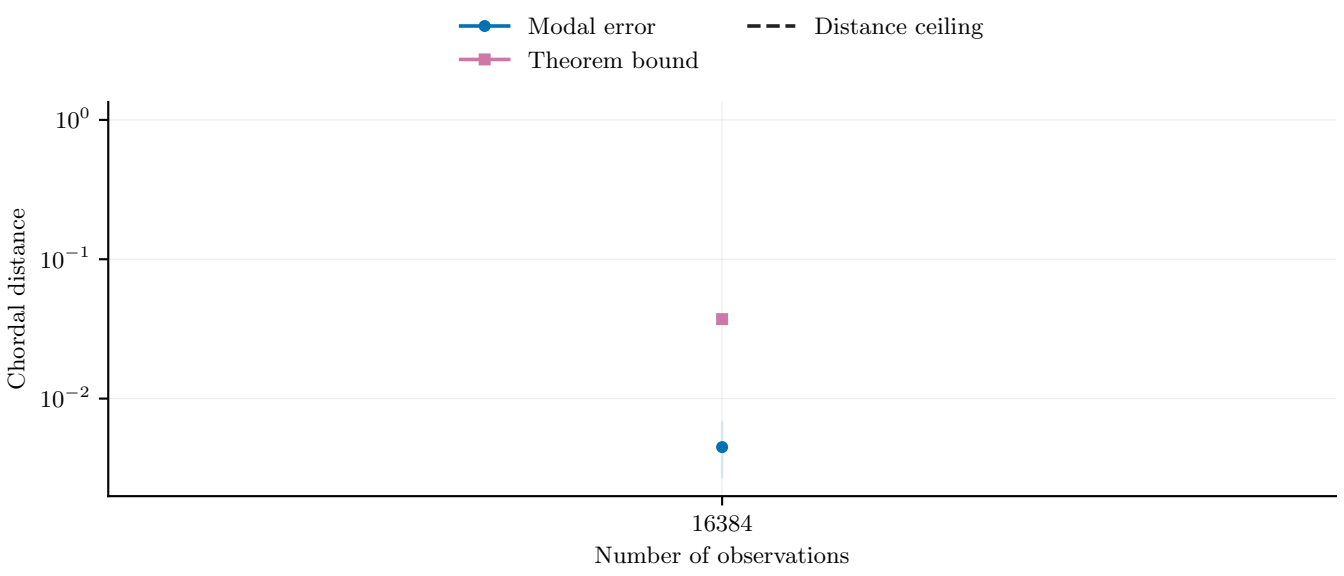
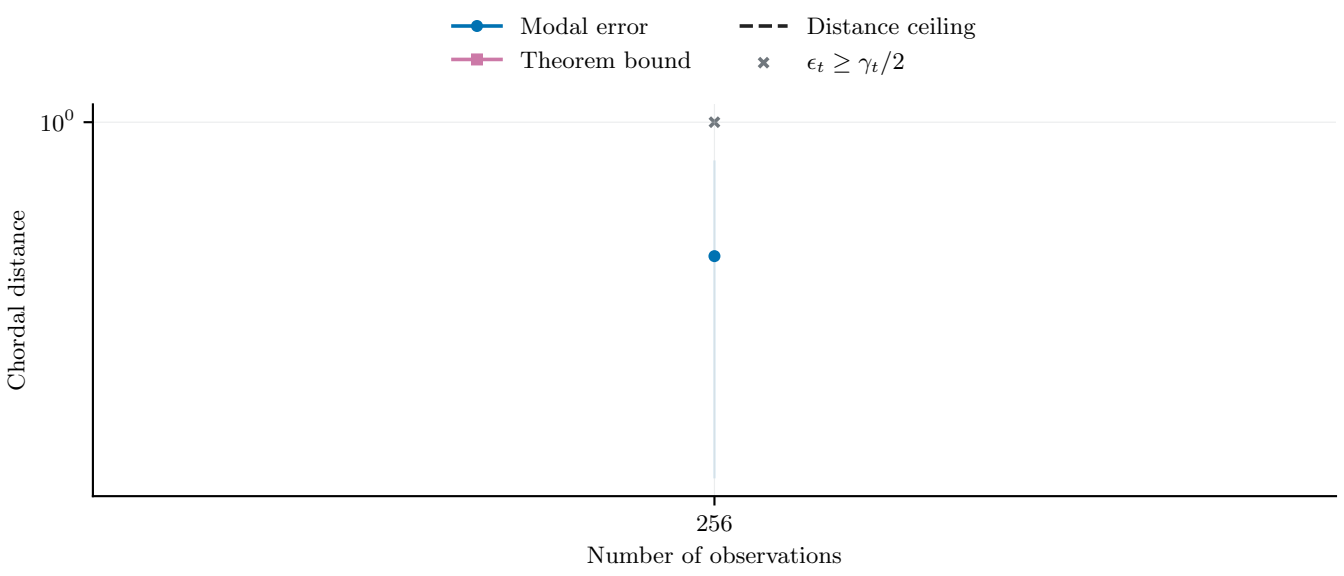

(a)
10^1
10^0
Estimated Fisher eigenvalue
10^0
10^1
Analytical Fisher eigenvalue
(b)
60
50
40
30
20
10
0
Fisher eigenvalue
1.0
1.5
2.0
Tangent direction


(a)
Empirical log-coordinate variance
10^{-1}
10^{-2}
10^{-3}
10^{-3}
10^{-2}
10^{-1}
H_A^{-1} tangent variance
Local regime
Outside local regime
(b)
Squared tangent error
Empirical MSE
$\mathrm{tr}(\mathcal{F}^{-1})$
10^{1}
10^{2}
10^{3}
Number of observations
(c)
Local fraction / normalized bias
1.0
0.8
0.6
0.4
0.2
0.0
Local fraction
Bias/$\sqrt{\mathrm{tr}(\mathcal{F}^{-1})}$
10^{1}
10^{2}
10^{3}
Number of observations


Modal error
Theorem bound
Distance ceiling
$\epsilon_t \geq \gamma_t/2$
(a)
Chordal distance
10^0
10^{-1}
10^{-2}
10^2
10^3
10^4
Number of observations
(b)
Operator perturbation
1000
800
600
400
200
200
400
600
800
1000
Theoretical allowance ϵ_t
(c)
Forgetting factor β
1
$1 - 10^{-4}$
0.999
0.995
0.99
0.98
0
1e-05
0.0001
0.001
Subspace angular speed
$\epsilon < \gamma/2$
$\epsilon \geq \gamma/2$


$\gamma_t/2$
Initialization
Unmodelled motion
Measurement noise
Operator perturbation allowance
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
Abrupt $\beta = 0.99$
Abrupt $\beta = 1$
Smooth $\beta = 0.99$
Smooth $\beta = 0.999$
Smooth $\beta = 1$
Static $\beta = 0.99$
Static $\beta = 1$
Concentrated inputs $\beta = 1$
Nonuniform prior $\beta = 1$


Modal error
Distance ceiling
Theorem bound
10^{0}
10^{-1}
10^{-2}
Chordal distance
16384
Number of observations


Modal error
Theorem bound
Distance ceiling
$\epsilon_t \geq \gamma_t/2$
10^0
Chordal distance
256
Number of observations